Kiyoung Om

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RESEARCH INTERESTS

- Generative Models for Optimization and Control
- Alignment of Generative Models via Search and Fine-tuning

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea — Mar 2024 - Feb 2026
Master of Science in Data Science — GPA: 3.78/4.3
Graduate School of Data Science
Advisor: Prof. Jinkyoo Park, System Intelligence Lab

Korea University, Seoul, South Korea — Mar 2018 - Feb 2024
Bachelor of Science in Industrial Management Engineering — GPA: 4.02/4.5, Major GPA: 4.19/4.5
Military Service: Republic of Korea Army — Sep 2019 - Feb 2021

PUBLICATIONS

Conference Papers & Workshop Proceedings

- Kang, H.*, Lee, J.*, Shin, W.*, **Om, K.**, & Park, J. (2026). Diffusion Fine-Tuning via Reparameterized Policy Gradient of the Soft Q-Function. *International Conference on Learning Representations (ICLR), Poster.* arXiv OpenReview (*Equal contribution)
- Lee, J., Kim, M., Choi, S., Song, I., Yun, S., Kang, H., Shin, W., Yun, T., **Om, K.**, & Park, J. (2026). Diffusion Alignment as Variational Expectation-Maximization. *International Conference on Learning Representations (ICLR), Poster.* arXiv OpenReview
- Yun, T.*, **Om, K.***, Lee, J., Yun, S., & Park, J. (2025). Posterior Inference with Diffusion Models for High-dimensional Black-box Optimization. *International Conference on Machine Learning (ICML); FPI @ ICLR Workshop.* arXiv OpenReview (*Equal contribution)
- **Om, K.***, Sim, K.*, Yun, T.*, Kang, H., & Park, J. (2025). Posterior Inference in Latent Space for Scalable Constrained Black-box Optimization. *NeurIPS Workshop on Structured Probabilistic Inference & Generative Modeling (Oral).* arXiv OpenReview (*Equal contribution)

Preprints

- Hwang, S.*, **Om, K.***, Kim, D., & Lee, J. (2026). Flow-ERD: Agent-type Aware Flow Matching with Entropy-Regularized Distillation for Diverse Traffic Simulation. *arXiv preprint arXiv:2607.06957.* arXiv Project (*Equal contribution; ranked 1st on the WOSAC 2025 Sim Agents Challenge)

Manuscripts & Industry White Papers

- Jung, J., Sim, Y., Lee, J. H., **Om, K.**, Shin, H., Kwon, C. (2025). *AI-accelerated Heuristics for Production Planning.* Samsung Electronics-KAIST industry white paper. Under review.

EXPERIENCE

NAVER LABS, Robotics & Embodied AI Team — Seongnam, South Korea
Research Intern — Mar 2026 - Present

- Researching learning-based traffic simulation for autonomous driving, with a focus on closed-loop covariate shift and realistic multi-agent behavior.
- Contributed to Flow-ERD, a flow matching based traffic simulator that ranked 1st on the WOSAC 2025 Sim Agents Challenge leaderboard.

PROJECTS

Samsung Electronics-KAIST AI Industry-Academia Collaboration Daejeon, South Korea
AI Research Assistant Mar 2025 - Sep 2025

- Developed an AI-accelerated Lagrangian heuristic for single-site CLSP with production leveling and joint capacity constraints.
- Predicted near-optimal Lagrangian multipliers to warm-start subgradient updates; achieved $\sim$75% runtime reduction vs. primal MILP and gap $< 0.1\%$, with $\sim$96% plan similarity to optimal.

Korea University Solar Cell Lab AI Industry-Academia Collaboration Seoul, South Korea
AI Application Advisor Apr 2023 - Jul 2023

- Developed prediction models for weighted-average reflection of solar cells using panel images (CNN).
- Created an experimental date and efficiency prediction system for Perovskite Solar Cells based on weather forecasting.

TEACHING EXPERIENCE

KAIST Industrial & Systems Engineering Daejeon, South Korea
Teaching Assistant - IE343 Statistical Machine Learning Mar 2024 - Jun 2024

- Supported undergraduate students in developing foundation in machine learning concepts.
- Graded assignments and provided detailed feedback on student work.
- Covered topics including supervised/unsupervised learning, regression, classification, and model evaluation.

Korea University Seoul, South Korea
Teaching Assistant - Deep Learning Programming Study Jul 2023 - Aug 2023

- Assisted students with implementing neural networks using PyTorch under Professor Sungbin Lim.
- Guided participants through practical exercises in deep learning.

AWARDS

Academic Excellence Award Korea University, 2022
Capstone Excellence Award Korea University, 2022
Topic: Understanding the Lifecycle and Gap Technologies of Lithium-Ion Batteries
Commander's Commendation Republic of Korea Army, 2020
Speech Contest on Practicing the Core Values of the Army and Showcasing Examples

EXTRACURRICULAR ACTIVITIES

KUBIG (Korea University BIG data research club) Seoul, South Korea
Regular Member Jul 2022 - Jul 2023

- Participated in and led study groups on NLP, Computer Vision, and Reinforcement Learning.
- Developed a vehicle hazard detection system using the YOLOv7 architecture.
- Led an algorithmic trading and backtesting research project using the FinRL framework.